Do vegetables and fruits reduce the risk of chronic obstructive pulmonary disease? A case-control study in Japan.

OBJECTIVE: To investigate the relationship between vegetable and fruit consumption and the risk of chronic obstructive pulmonary disease (COPD), a case-control study was conducted in central Japan in 2006. METHODS: A total of 278 referred patients with COPD diagnosed within the past four years and 340 community-based controls undertook spirometric measurements of respiratory function. A structured questionnaire was administered face-to-face to obtain information on demographics, lifestyle and habitual food consumption. RESULTS: The mean vegetable and fruit intakes of cases (155.62 (SD 88.84) and 248.32 (SD 188.17) g/day) were significantly lower ($p<0.01$) than controls (199.14 (SD 121.41) and 304.09 (SD 253.72) g/day). A substantial reduction in COPD risk was found by increasing daily total vegetable intake, p for trend=0.037. The prevalence of breathlessness also decreased with vegetable consumption, the adjusted odds ratio being 0.49 (95% CI 0.27-0.88) for the highest versus lowest quartile of intake. However, the effects of fruit consumption were not significant. Among the nutrients contained in vegetables and fruits, vitamin A was particularly significant ($p=0.008$) with an estimated 52% reduction in COPD risk at the highest level of intake. CONCLUSION: The study provided evidence of an inverse association between vegetable consumption and the risk of COPD for Japanese adults.